

BILL OF MATERIALS

MastitisGuard AI — Smart Dairy Health Monitoring and Early Mastitis Detection System

Team Name: H14_SPIRIT

Team ID: 146962

Problem Statement Number: 26109

Journey stage: Built solution

1. Overview

This document itemizes the components required to build the **three-unit MastitisGuard AI prototype: the Animal Wearable Collar, the Milk Measurement Unit, and the Shed Edge-AI Unit**. Costs are estimated at single-unit prototype quantities, sourced from standard Indian electronics retailers; per-unit cost falls substantially at production scale (see the Market Research document, Section 6, for the volume-pricing view — the per-cow figure quoted there, ₹2,510, is drawn directly from this document).

2. Animal Wearable Collar — ₹2,510 (prototype grade)

The wearable is a 3D-printed, IP65-equivalent collar unit worn continuously by the animal. Certified IP69K injection-moulded housing is a Phase 2 production target (Section 6), not the current build.

Component	Purpose	Cost
ESP32 DevKit V1	Core processing + BLE/Wi-Fi	₹280
MAX30205 clinical-grade contact temperature IC	Skin-contact temperature sensing	₹180
MPU6050 6-axis IMU	Accelerometer + gyroscope — activity/rumination sensing	₹130
NEO-6M GPS module	Pasture mobility tracking	₹280
18650 Li-ion cell, 3.7V 2,600 mAh	Primary power storage	₹310
TP4056 charge management IC	Battery charge management	₹35
3D-printed IP65-equivalent enclosure + PCB	Weatherproof housing (uncertified prototype)	₹500
Woven nylon collar strap + buckle	Attachment to animal	₹645
Wiring, connectors, assembly	Assembly consumables	₹150

Subtotal — Wearable Collar (prototype grade): ₹2,510

Optional field variant (open-pasture grazing, Phase 2): add a **curved monocrystalline solar panel (₹450)** and a **LoRaWAN/BLE sub-GHz module for long-range telemetry (₹420)** — Extended Wearable Total: **₹3,380**. The core ₹2,510 build is unaffected; these are additive, not a replacement of any component above.

3. Milk Measurement Unit

Installed at the milking point; identifies the animal and records milk-quality parameters at each session. Two build tiers exist, at two different points in the scaling plan (Market Research document, Section 4):

3.1 Prototype Grade — ₹2,110 (current build)

Component	Purpose	Status	Cost
ESP32 DevKit V1	Milk-unit controller	Built & bench-tested	₹320
DS18B20 waterproof probe	Milk temperature / pH thermal compensation	Built & bench-tested	₹120
HX711 load cell amplifier	Weight signal amplification	Built & bench-tested	₹90
S-type/D-ring load cell (5–20 kg)	Milk yield measurement	Built & bench-tested	₹280
PHSENS pH sensor module + BNC probe	Milk pH	Budgeted; physical integration is Phase 1 pilot task	₹650
ESP32-CAM + OV2640	QR/ID scan; CMT vision planned	ID scan tested; CMT vision module not yet built	₹450
Wiring, connectors, waterproof enclosure	Assembly consumables	Built	₹200

Subtotal — Milk Unit, prototype grade: ₹2,110

3.2 Field Grade — ₹5,250 (Phase 2 scale-up target)

The field-grade build starts from the prototype-grade ₹2,110 above and adds the upgrades needed for continuous multi-cow field operation:

Upgrade over prototype grade	Cost
Prototype-grade base (Section 3.1)	₹2,110
Food-grade 316 stainless-steel fluidic housing & mounting bracketry	₹1,200
Automated sample chamber & dosing tray (consistent pH sampling across sessions)	₹950
Environmental sealing, calibration, and field assembly labour	₹990
Total — Field Grade	₹5,250

4. Shed Edge-AI Unit — ₹950

The central processing node — receives wearable and milk-unit data over ESP-NOW and runs the offline baseline + CatBoost inference engine described in the ML Model Documentation.

Component	Purpose	Cost
ESP32 DevKit V1	Central Edge-AI unit — runs offline baseline + CatBoost model	₹320
OLED display (SSD1306, optional)	On-site risk display	₹180
5V power supply / battery backup	Continuous shed-side power	₹250
Enclosure + mounting hardware	Protects central unit in shed environment	₹200

Subtotal — Shed Unit: ₹950

5. Total Prototype Cost (Per Cow / Per Farm Unit)

A. Base Functional Prototype Setup (1 Cow + Shared Farm Infrastructure)

Line item	Cost
Wearable Collar Unit (prototype grade)	₹2,510
Milk Measurement Unit (prototype grade)	₹2,110
Shed Edge-AI Unit	₹950
Total Base Starter Cost (1 cow + 1 farm setup)	₹5,570

B. Shared Farm Station Subtotal (Per Farm, Not Per Cow)

Milk Measurement Unit (₹2,110) + Shed Edge-AI Unit (₹950) = ₹3,060. Field-grade Phase 2 upgrade to the Milk Unit raises this to ₹6,200 (₹5,250 + ₹950) — see Section 3.2 for the exact composition of that upgrade.

C. Extended Field Setup (With Solar & LoRaWAN Add-Ons)

Wearable Collar Unit with add-ons: ₹3,380 (₹2,510 + ₹450 solar panel + ₹420 LoRaWAN module). Total Extended Prototype (1 extended wearable + prototype-grade milk unit + shed unit): ₹6,440 (₹3,380 + ₹2,110 + ₹950).

D. Incremental Cost for Additional Cows

Because the Shed Unit (₹950) and Milk Unit (₹2,110 prototype / ₹5,250 field-grade) are shared across the herd, scaling to additional animals requires only individual wearable collars:

- Base prototype: ₹2,510 per additional cow.
- Extended setup (solar + LoRa): ₹3,380 per additional cow.

E. Initial 10-Cow Field Pilot Budget

Line item	Cost
10 Wearable Collars (prototype grade)	₹25,100
1 Milk Measurement Unit (prototype grade)	₹2,110
1 Shed Edge-AI Unit	₹950
Spares / testing / contingency (~15%)	₹4,240
Estimated Total	₹32,400

Indicative prototype budget: ₹30,000–₹35,000. Per-cow incremental hardware cost (₹2,510) is an order of magnitude below the global USD 50–400 (≈ ₹4,000–₹34,000) industry entry band — the same comparison made in the Market Research document, Section 6.

6. Sourcing Matrix (Prototype vs. Scale) & Build Status Disclosure

Procurement Tier: All pricing above represents single-unit retail component costs (Indian domestic electronics distributors) for prototype assembly; bulk component purchasing (100+ units) reduces bill-of-materials cost significantly.

Build Status Disclosure:

- Operational on prototype today: ESP32 DevKit cores, MPU6050 IMU, DS18B20 milk-temperature probe, HX711 + load cell, TP4056 PMIC, 18650 Li-ion cells, NEO-6M GPS. Skin-contact temperature is currently bench-tested with a DHT11 as a temporary stand-in; the costed and specified component for the pilot build is the MAX30205 (Section 2).
- Budgeted for pilot phase, not yet physically integrated: the PHSENS pH glass electrode module and the ESP32-CAM optical ear-tag scanner/CMT vision unit. Both are priced above and tested in firmware/data structures; physical integration into the milking manifold is the defined Phase 1 pilot task (see Decision-Making and PoC document, Section 4).
- Mechanical enclosures: Current prototype uses custom FDM/SLA 3D-printed and CNC-milled enclosures, rated to an IP65-equivalent standard but not lab-certified. Production units will shift to high-volume injection-moulded, UV-stabilized ABS/polycarbonate housing certified to IP69K, reducing per-unit housing cost at scale (Market Research document, Section 4, Phase 2).

7. References

- Espressif Systems. ESP32-C3 Series Datasheet. Espressif Systems, Shanghai. espressif.com/en/products/socs/esp32-c3.
- International Organization for Standardization. ISO 20653:2013 — Road vehicles: Degrees of protection (IP code) — basis for the IP65/IP69K ingress-protection ratings referenced for the wearable collar housing.